

Maggie Makar

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Education	Massachusetts Institute of Technology Ph.D, Electrical Engineering and Computer Science, Expected May 2020 University of Massachusetts, Amherst B.S., Mathematics and Economics, <i>Summa Cum Laude</i> , Feb 2013
Computer and Language Skills	<u>Computing:</u> R, Python, Julia, L ^A T _E X, and Stata <u>Languages:</u> English, Arabic, and French
Current research	I study machine learning, and applications of machine learning in important domains, primarily human health. I'm interested in studying network models, causality, and developing accurate, scalable inference methods for Bayesian models.
Research Experience	Graduate Research Assistant , Computer Science and Artificial Intelligence Laboratory, MIT, Cambridge, MA, Sept 2015 - Present Working in the Data driven inference group. Studying information diffusion over graphs and social network models. Creating models to understand and interpret the causal mechanisms governing the spread of information and contagions. Developing approximate methods to do inference over large graphs. Research Assistant , Brigham and Women's Hospital, Boston, MA, Feb 2013 - August 2015 Developed and applied machine learning methods to discover patterns of healthcare utilization and disease progression within older adults and the terminally ill. Focused on Medicare patients with metastatic cancers. Developed visualization tools to summarize important population-wide health metrics. Studied model evaluation in the presence of extremely rare outcomes. Research Assistant , Political Economy Research Institute, Amherst, MA Sept - Dec 2012 Used statistical methods to analyze and compare energy policies of 5 countries as a part of a UN commissioned project with Professor Robert Pollin. Research Intern , Donahue Institute, Hadley, MA, May - Aug 2012 Built statistical models to study economic topics such as the effect of the defense industry's investments on the New England Economy. Research Assistant , Center for Intelligent Information Retrieval, Amherst, MA Feb -May 2012 Tested and validated a cross-lingual information retrieval system for informal document genres. Contributed to developing an Egyptian morphological stemmer Undergraduate Research Project , Economics Department, University of Massachusetts, Amherst, Fall 2012 "Deregulation of the Electricity Market and its Effect on Nuclear Power Plants in the U.S: A critical replication" Used impulse response functions to study the effect of electricity market deregulation on the efficiency of nuclear power plants. Analyzed monthly data collected from 103 U.S based nuclear power plants.

Undergraduate Research Assistant, Math Department, University of Massachusetts, Amherst, **Fall 2012**

"Nanoscale Topography in a Beaker : Characterization and Development of Patterned Si Surfaces"

Measured and analyzed the effect of chemical manipulation on surface topography using hierarchical linear models.

Manuscripts in Preparation

Makar, M., Shenoy, E., Gutttag, J., Wiens, J., "Probability of Activation in the Presence of Latent Spreaders: Identifying hidden influencers in social networks"

Publications

Obermeyer, Z., **Makar, M.**, Abujaber, S., Dominici, F., Block, S., Cutler, D., "Association between the Medicare hospice benefit and health care utilization and costs for patients with poor-prognosis cancer". *JAMA*. 2014 Nov 12; 312(18): 1888-1896

Makar, M., Ghassemi, M., Cutler, D., Obermeyer, Z., "Short-term mortality prediction for elderly patients using Medicare claims data". *International Journal of Machine Learning and Computing*. 2015 Vol. 5(3): 192-197

Powers, B., **Makar, M.**, Cutler, D., Obermeyer, Z., "Cost Savings Associated with Expanded Hospice Use Among Medicare Beneficiaries with Poor Prognosis Cancer". *J Palliat Med*. 2015 May; 18(5):400-1

Obermeyer, Z., Abujaber, S., **Makar, M.**, Stoll, S., Kayden, S., Wallis, L., Reynolds, T., "Emergency care delivery in 59 low- and middle-income countries: Systematic review". *WHO Bulletin*. May 2015

Obermeyer, Z., Powers, B., **Makar, M.**, Keating, N., Cutler, D., "Physician revealed preferences for hospice are strongly related to their patients' enrollment in hospice". *Health Affairs*. 2015 Jun 1; 34(6):993-1000

Obermeyer, Z., Clarke, A., **Makar, M.**, Schuur, J., Cutler, D., "Emergency Care Use and the Medicare Hospice Benefit for Individuals with Cancer with a Poor Prognosis". *J Am Geriatr Soc*. 2016 Feb; 64(2):323-9

Submitted Manuscripts

Makar, M., Antonelli, J., Di, Q., Schwartz, J., Cutler, D., Dominici, F., "Estimating the Causal Effect of Lowering Particulate Matter Levels Below the National Ambient Air Quality Standards on Health Outcomes"

Obermeyer, Z., **Makar, M.**, Freedman, A., Temel, J., Mullainathan, S. "Improving prognosis: Development of a novel method for prediction of six-month mortality in a cohort of Medicare beneficiaries".

Joynt, K., **Makar, M.**, Gwande, A., Cutler, D., Obermeyer, Z., "Understanding Epidemiology and Costs for the Highest-Cost Medicare Inpatients"

Obermeyer, Z., Powers, B., **Makar, M.**, Bohlen, M., Cutler, D., "The effect of hospice on out-of-pocket expenditures at the end of life"

Obermeyer, Z., **Makar, M.**, Jena, A., Cutler, D. "Unexpected death after discharge home from the emergency departments in elderly Medicare beneficiaries."

**Presentations,
talks and poster
sessions**

Inference on graphs with missing nodes
Invited talk at the University of Michigan in Ann Arbor, July 2016

Estimating the Causal Effect of Lowering Particulate Matter Levels Below the National Ambient Air Quality Standards on Health Outcomes
Accepted for presentation at the 2016 Joint Statistical Meeting (JSM), Chicago

Learning meaningful representations from medical claims data
Accepted for presentation at the machine learning for healthcare workshop at the twenty-ninth annual conference on neural information processing systems (NIPS), December 2015

Short-term mortality prediction for elderly patients using Medicare claims data.
Accepted for presentation at the Machine Learning summer school, Max Planck Institute, Tuebingen, July 2015

Association between the Medicare hospice benefit and health care utilization and costs for patients with poor-prognosis cancer
Invited talk at Harvard School of Public Health, Statistical Informatics for Cancer Research Group, October 2014

Improving prognosis: Development of a novel method for prediction of six-month mortality in a cohort of Medicare beneficiaries.
Accepted for presentation at the 2014 ninth annual Women in Machine Learning Workshop, December 2014

Short-term mortality prediction for elderly patients using Medicare claims data.
Accepted for presentation at the 2015 7th International Conference on Machine Learning and Computing, March 2015

Teaching

Co-organizer, instructor: Whirlwind of Machine learning. Winter (IAP) 2017. MIT
Teaching assistant: Machine learning for Healthcare. Spring 2017. MIT

Awards

Ward McCarthy scholarship
School of Behavioral and Social Sciences scholarship
Economics Department Writing Award
Life Members Scholarship

**Applicable
Graduate Courses**

- Machine learning
- Bayesian Modelling
- Optimization
- Computer Networks
- Cryptography and cryptanalysis

Other Experience

Consultant, Cyft Healthcare Analytics
Intern, International Development Law Organization
Junior Analyst, Makary Consulting, Cairo, Egypt

Sept 2016-Present
Jan 2012-June 2012
Summer 2011